

Tab 1

# Syllabus Repository, Revision and Versioning System (SRVS)

## Software Requirements Specification (SRS)

TUQUIB, GIAN CARLO  
LIMA, RALPH JAMES  
MIRAVELES, ALSERGE VINCENT

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### PART 1: USER KEY ROLES

The system has exactly 4 roles. Each role has its own set of permissions, its own dashboard, and a specific position in the syllabus approval workflow. Roles are assigned by the System Administrator after a mandatory user registration process. A user can only hold one role at a time.

#### ROLE 1: System Administrator

- **Who they are:** The superuser of the platform's infrastructure and security layer. The Admin has no visibility or access to the syllabus repository, document contents, or approval queues. They focus exclusively on user access, account states, and system monitoring.
- **What they can do:**
  - **User Account Management (Registration Queue):**
    - Access the unified User Approval Queue showing all registered accounts (Admins, Department Heads, Educators, and Students) pending institutional verification.
    - Approve pending registrations to transition accounts to active status, which triggers an automated system notification.
    - Reject pending registrations with an optional reason string, preventing system access.

- Activate, deactivate, or delete user accounts at any time. Deactivated accounts lose all login privileges, but their historical data records and audit logs remain completely preserved.
    - Search any registered user across the institution by name, email, or institutional ID.
    - View a user's comprehensive session history, role tracking metadata, and platform activity logs.
  - **System Monitoring and Audit Trail:**
    - Access the complete, unfiltered system-wide audit trail capturing every operational transaction across the environment.
    - Execute queries and apply filters to audit logs by Action Type, User Identity, Result Status, or UTC Date Range.
  - **What they CANNOT do (Explicit Restrictions):**
    - Cannot view, read, search, or download any syllabus PDF stored in the repository.
    - Cannot access the Pending Reviews Queue, approve, or reject any syllabus submission.
    - Cannot receive notifications regarding syllabus workflow events (submissions, re-uploads, or rejections).
    - Admin no longer accesses repository/review queues.
  - **Admin Dashboard Contents:**
    - **System Statistics Panel:** Metric cards displaying Registered User Count and Total Files Stored (excluding content details).
    - **User Approval Queue:** A data table managing all pending registrations across all 4 system roles.
    - **Recent Activity Feed:** A real-time stream showcasing the last 10 system-wide account or configuration events.
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## ROLE 2: Department Head / Academic Coordinator

- **Who they are:** The academic quality-control and primary approval layer. Each Department Head is assigned to one or more academic departments. They ensure syllabi meet curriculum standards. They possess full syllabus PDF upload and final approval authority for their assigned departments.
- **What they can do:**
  - **Syllabus PDF Upload:**
    - Upload a new syllabus PDF for any course within their assigned department(s) along with indexing metadata (Course Code, Course Title, Year/Semester).
  - **Workflow Review and Final Approval Authority:**
    - Access the departmental Pending Reviews Queue displaying all syllabi submitted by Educators within their assigned department(s).

- Review submitted documents using an integrated web-based PDF viewer or via direct download.
    - **APPROVE** a submission: Updates the syllabus status to *Approved*, immediately publishes the file to the student viewer portal, and notifies the Educator.
    - **REJECT** a submission: Updates the status to *Rejected*, records mandatory written feedback detailing required modifications, notifies the Educator, and reverts the file to an editable *Draft* state.
  - **Curriculum Oversight & History Monitoring:**
    - View all syllabi within their assigned department(s) across all lifecycle phases (Draft, Submitted, Approved, Rejected).
    - Access the file version history table for any course to track previous file changes and download older iterations of the PDF.
  - **What they CANNOT do (Explicit Restrictions):**
    - Cannot view or access syllabi, audit entries, or files outside of their assigned department(s).
    - Cannot approve, reject, or manage user accounts or modify security roles.
  - **Dept. Head Dashboard Contents:**
    - **Department Overview Panel:** All courses under their purview mapped to active syllabus statuses.
    - **Pending Reviews Queue:** Operational workflow list displaying incoming syllabus PDF submissions with inline Approve and Reject actions.
    - **Recent Upload Activity Feed:** Chronological change stream tracking all departmental syllabus file updates.
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### ROLE 3: Educator (Instructor / Faculty Member)

- **Who they are:** Primary content providers who upload and maintain syllabi for the specific courses they are assigned to instruct. All PDF files they upload must pass through the workflow layer for Department Head approval before student publication.
- **What they can do:**
  - **Syllabus File Upload:**
    - Upload a finalized syllabus PDF for an assigned course using a streamlined indexing form (Course Code, Title, Department, Academic Year, Semester).
    - Save the file as a Draft, allowing them to replace the PDF as needed before formal submission.
  - **Version Control Management:**
    - View a simplified file-history log of their course syllabus showing all previously uploaded PDFs, timestamps, and change text notes.

- Download previous versions directly from the history log to recover old document contents.
  - **Workflow Submission:**
    - Route a completed draft to the Department Head's Pending Reviews Queue via a Submit action, changing its status from *Draft* to *Submitted*.
    - Receive notifications when the Department Head processes the submission with an Approved or Rejected decision.
  - **Educator Dashboard Contents:**
    - **My Syllabi Tracker:** Interactive tiles breaking down owned documents by status (Draft, Submitted, Approved, Rejected).
    - **Notification Center:** Real-time alert feed surfacing workflow status updates and rejection notes.
    - **Quick-Action Buttons:** Upload New Syllabus PDF, View My Course Folders.
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## ROLE 4: Viewer (Student / Guest User)

- **Who they are:** End consumers of the approved curriculum data. They interact with the platform through a locked, read-only interface and can never see documents currently in flux.
  - **What they can do:**
    - **Browsing and Querying:**
      - Access a dedicated Viewer Dashboard displaying active, published course syllabi organized by department.
      - Query the approved repository using specific search parameters: Course Name, Course Code, Department, or Academic Year/Semester.
    - **Viewing and Downloading:**
      - Open and read approved syllabus files inside an integrated online PDF viewer.
      - Download or print the published PDF file for offline utility.
    - **What they CANNOT do (Explicit Restrictions):**
      - Cannot upload, submit, or alter any file records.
      - Cannot see files residing in *Draft*, *Submitted*, or *Rejected* states under any circumstances.
      - Cannot access the historical version table, upload logs, or administrative views.
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## PART 2: GENERAL FUNCTIONS

### FUNCTION 1: Role-Based Login and Authentication

- **Universal Registration Workflow:**
    - Every user—including prospective Admins, Department Heads, Educators, and Students—must onboard via the public registration interface (`/register`). Direct manual account creation inside dashboards is prohibited.
    - Upon submitting the form, the account enters a **Pending Approval** state and is locked from all system features until the Admin approves it from the queue.
  - **Strict Institutional ID Validation Rules:**
    - The registration form strictly validates institutional identification number parameters. Accounts failing these structural patterns are rejected instantly:
      - **Department Head ID:** Must consist of **exactly 5 digits**.
      - **Educator / Faculty ID:** Must consist of **exactly 5 digits**.
      - **Student / Viewer ID:** Must consist of **exactly 10 digits**.
  - **Authentication Processing:**
    - Users authenticate at `/login` via their registered email and password. Valid authentication triggers a role-specific dashboard routing redirection.
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## FUNCTION 2: Syllabus Upload and Metadata Indexing

- **Purpose:** Replaces form-based document creation with a strict file ingestion pipeline.
  - **Behavior:**
    - Instead of writing the syllabus within a text editor on the website, the user attaches a finalized `.pdf` file.
    - To make the file searchable, the uploader must fill out a minimal indexing form containing:
      - *Index Metadata:* Course Code, Course Title, Department, Academic Year, and Semester/Term.
    - When the file is saved, it is written directly to a secure system file storage directory, and the metadata record is stored in the database with an initial status of `Draft`.
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## FUNCTION 3: Simplified File-Based Versioning System

- **Purpose:** Provides a straightforward, highly plausible approach to tracking changes without database snapshots or programmatic text diffing.
- **Core Architecture:**
  - Versioning is managed by **file replacement and file-naming serialization**.
  - The server maintains a file directory or bucket. Each uploaded file is saved using a structured tracking convention:  
`[CourseCode]_[Semester]_[VersionNumber].pdf` (e.g.,  
`CS301_FirstSem2026_V1.pdf`).

- When an Educator edits a draft or revises a rejected syllabus, they do so by uploading an updated PDF file.
  - **The Overwrite/Increment Process:**
    - To save an update, the system forces the user to fill out a mandatory text field: **Change Summary** (e.g., *"Updated grading breakdown in week 5 as requested by the Dept Head"*).
    - Upon saving, the system leaves the older file (**\_V1.pdf**) completely untouched in storage, saves the new file as **\_V2.pdf**, and logs a new row in the version history table.
  - **The Version History Table UI:**
    - Authorized users see a clean chronological log table containing:
      1. **Version Target:** (e.g., V1, V2, V3)
      2. **Upload Timestamp:** UTC Date and Time.
      3. **Actor Identity:** The name of the user who uploaded the file.
      4. **Change Summary Text:** The written description of what was altered in this revision.
      5. **Action Button:** A direct link to **Download PDF** for that specific historical version.
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## FUNCTION 4: Simplified Version Review and Comparison

- **Purpose:** Replaces programmatic side-by-side text highlighting (red/green text diffs) with an easily implementable review workflow.
  - **Review Behavior:**
    - When a Department Head opens a syllabus from their Pending Reviews Queue, they are presented with the current active PDF via an embedded web viewer module.
    - Next to the document viewer, a **File Version History Panel** displays the table of past uploads.
    - **Comparison Method:** To see what changed, the Department Head reviews the **Change Summary Text** provided by the educator for the latest version. If they need to double-check the exact file differences, they can click the download action next to any previous version (e.g., V1) to view it alongside the new submission (V2) manually. This removes the need for complex server-side PDF text parsing.
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## FUNCTION 5: Simplified Version Restoration

- **Purpose:** Allows an author to easily roll back to a previous file iteration.
- **Process Flow:**

1. The user navigates to the Version History Table for a course.
  2. They select a past file entry (e.g., V1) and click a **Promote to Current Draft** button.
  3. The system takes a copy of the V1 PDF file in storage, duplicates it, and saves it as the highest incremental version number (e.g., V3) with a system-generated change summary: *"Restored content from Version 1"*.
  4. This newly generated version initializes in an editable **Draft** status, preserving all previous version histories completely.
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## FUNCTION 6: Syllabus Submission and Review Workflow

- **Purpose:** Ensures no syllabus PDF becomes visible to students until it clears departmental review.
  - **The 4 Workflow Status States:**
    1. **Draft:** The file is mutable. The Educator can continuously replace the PDF file. Version logs increment with each save. Hidden from students.
    2. **Submitted:** The Educator clicks "Submit". The file is **locked from editing**. It populates the Department Head's queue.
    3. **Approved:** The Department Head approves the file. It immediately becomes visible on the student search dashboard.
    4. **Rejected:** The Department Head rejects the file and logs feedback. The status reverts to **Draft**, unlocking the upload button for the Educator to submit a revised PDF (which will become the next version entry).
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## FUNCTION 7: Scoped Repository Search

- **Behavior:**
    - **Students:** Can query the system by Course Code or Title, but the database filter returns hits *only* where **Status = Approved**. They receive a direct download link for the approved PDF.
    - **Educators / Department Heads:** Can search all files matching their assigned departments across all status filters (**Draft**, **Submitted**, **Approved**, **Rejected**) to manage workflows.
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## FUNCTION 8: Activity Logging and Audit Trail

- **Purpose:** Maintain an append-only, immutable record of system events for security and transparency.



- **Monitored Upload Actions:**

- The system logs an entry every time a file is uploaded, submitted, approved, rejected, or restored.
  - *Example Log Entry:* [Timestamp] - User [Faculty ID: 12345] uploaded a new file version V2 for Course [CS301] with status [Submitted].
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## FUNCTION 9: Notification System

- **Behavior:** Automatically handles targeted alerts regarding file updates:
    - *Submission Alert:* Notifies the Department Head when an Educator submits a PDF.
    - *Approval/Rejection Alert:* Notifies the Educator immediately when a decision is logged, packing the Department Head's written notes inside rejection alerts.
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Tab 2

# **SRVS — USER KEY ROLES & GENERAL FUNCTIONS**

## **Syllabus Repository, Revision, and Versioning System**

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### **PART 1 — USER KEY ROLES**

The system has exactly four (4) roles. Each role has its own set of permissions, dashboard, and responsibilities within the syllabus submission and approval workflow. Roles are assigned after registration approval. A user can only hold one role at a time.

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### **ROLE 1 — System Administrator**

#### **Who they are**

The System Administrator is responsible for account management, registration approval, system monitoring, audit tracking, and overall platform maintenance. Administrators do not participate in syllabus approval or syllabus repository management.

#### **What they can do**

##### **User Account Management**

- View all registered users in the system
- Approve or reject newly registered accounts
- Activate or deactivate user accounts
- Assign or update user roles
- Search users by name, email, or ID number
- View user login history and account activity
- Permanently delete inactive accounts when necessary

##### **Registration Approval**

- Review pending registrations submitted by Faculty, Department Heads, and Students
- Approve registration requests
- Reject registration requests with optional remarks
- Notify users regarding account approval or rejection

## **System Monitoring and Audit**

- Access the complete audit trail of system activities
- Monitor login attempts, uploads, approvals, rejections, and account activities
- Filter audit logs by:
  - User
  - Action type
  - Date
  - Status
- Generate system usage reports and statistics
- View overall platform statistics including:
  - Total registered users
  - Total uploaded syllabi
  - Total processed syllabus records
  - Rejected syllabi count
  - Recent system activities

## **What they CANNOT do**

- Cannot upload syllabi
- Cannot edit syllabus files
- Cannot approve or reject syllabus submissions
- Cannot access syllabus review queues
- Cannot receive syllabus submission notifications
- Cannot modify syllabus version history

## **Admin Dashboard Contents**

- User management panel
- Pending registration approvals
- Audit logs and monitoring tools
- System statistics panel
- Recent activity feed
- Quick links:
  - Manage Users
  - Audit Logs
  - System Reports
  - Account Requests

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# ROLE 2 — Department Head / Academic Coordinator

## Who they are

Department Heads serve as the academic review and approval authority within their assigned department. They monitor syllabus compliance, review uploaded syllabi, and approve or reject submissions before publication.

## What they can do

### Syllabus Upload and Management

- Upload syllabus files for courses within their assigned department
- Enter syllabus metadata including:
  - Course Code
  - Course Title
  - Department
  - Academic Year
  - Semester
  - Instructor Name
- Edit and update their uploaded syllabi
- Submit uploaded syllabi for review

### Syllabus Review and Approval

- View all syllabi within their assigned department
- Review submitted syllabi uploaded by faculty members
- Approve syllabus submissions
- Reject syllabus submissions with feedback remarks
- Monitor syllabus compliance within the department
- View syllabus status including:
  - Draft
  - Submitted
  - Approved
  - Rejected

### Revision Tracking

- View uploaded syllabus revision history
- Access archived previous uploads of a syllabus
- View revision metadata including:
  - Version number
  - Upload date
  - Uploaded by
  - Revision note

## **Reporting and Monitoring**

- Generate department-level syllabus reports
- Monitor courses with missing syllabi
- Monitor recently updated syllabi
- Access audit entries related only to their department

## **What they CANNOT do**

- Cannot manage user accounts
- Cannot change user roles
- Cannot access audit logs outside their department
- Cannot access system-wide administrative settings

## **Department Head Dashboard Contents**

- Department syllabus overview
- Pending syllabus review queue
- Approved and rejected syllabus statistics
- Recent syllabus uploads
- Revision history panel
- Quick links:
  - Review Submissions
  - Upload Syllabus
  - Department Reports
  - View Archived Versions

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# **ROLE 3 — Faculty Member / Educator**

## **Who they are**

Faculty Members are responsible for uploading, updating, and maintaining syllabi for their assigned courses.

## **What they can do**

### **Syllabus Upload**

- Upload syllabus files for assigned courses
- Enter syllabus metadata including:
  - Course Code
  - Course Title
  - Department
  - Academic Year
  - Semester
  - Instructor Name
- Upload supported file formats such as:
  - PDF
  - DOCX

### **Revision Upload**

- Upload revised versions of previously submitted syllabi
- Add optional revision notes describing updates made
- View upload history of their own syllabi
- Access archived previous uploads

### **Submission Workflow**

- Save uploads as Draft
- Submit syllabi for Department Head review
- Receive notifications when submissions are:
  - Approved
  - Rejected
- Revise and resubmit rejected syllabi

### **Monitoring**

- View status of uploaded syllabi
- View upload timestamps and version numbers
- Monitor submission progress

## **What they CANNOT do**

- Cannot approve or reject syllabi
- Cannot access other departments' syllabi
- Cannot manage user accounts
- Cannot access audit logs
- Cannot delete approved syllabi

## **Faculty Dashboard Contents**

- My Uploaded Syllabi
  - Upload Status Overview
  - Recent Uploads
  - Revision History
  - Notification Center
  - Quick links:
    - Upload Syllabus
    - View Submissions
    - Upload Revision
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## **ROLE 4 — Student / Viewer**

### **Who they are**

Students are the end users of the system. They can only access approved syllabi.

### **What they can do**

#### **Browsing and Searching**

- Browse approved syllabi by department
- Search approved syllabi using:
  - Course Code
  - Course Title
  - Department
  - Semester
  - Academic Year

#### **Viewing and Downloading**

- View approved syllabus files using an embedded iFrame PDF viewer



- Printing and downloading of syllabus files are disabled
- Students are limited to read-only access

## **What they CANNOT do**

- Cannot upload syllabi
- Cannot edit syllabus files
- Cannot approve or reject submissions
- Cannot access draft or rejected syllabi
- Cannot access revision archives
- Cannot access audit logs

## **Student Dashboard Contents**

- Approved syllabus repository
  - Search bar
  - Department navigation panel
  - Download and print options
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# **PART 2 — GENERAL FUNCTIONS**

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## **FUNCTION 1 — Role-Based Login and Authentication**

### **Purpose**

Provides secure system access for all registered users.

### **Registration Rules**

All users must register before accessing the system.

## **ID Number Validation**

- Faculty Members and Department Heads must use exactly five (5) digits
- Students must use exactly ten (10) digits

## **Registration Fields**

- Full Name
- Email Address
- ID Number
- Password
- Role Selection
- Department

## **Registration Approval Flow**

1. User completes registration form
2. Account enters Pending status
3. Admin reviews registration request
4. Admin approves or rejects the account
5. Approved users may log in

## **Login Process**

1. User enters registered email and password
2. System validates credentials
3. Valid login redirects user to role-specific dashboard
4. Invalid login displays error message
5. Login attempts are recorded in audit logs

## **Forgot Password Process**

1. User clicks "Forgot Password"
2. User enters registered email address
3. System sends password reset link or verification code to the user's email
4. User creates a new password
5. Previous password becomes invalid

## **Session Security**

- Sessions expire after inactivity
- JWT stored securely using HTTP-only cookies

- CSRF protection enabled
  - Authentication activities logged in audit trail
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## **FUNCTION 2 — Role-Specific Dashboards**

Each role receives a personalized dashboard displaying only role-relevant features and data.

### **Admin Dashboard**

- User monitoring
- Pending registrations
- Audit logs
- System reports

### **Department Head Dashboard**

- Pending syllabus reviews
- Department syllabus monitoring
- Revision archives
- Approval statistics

### **Faculty Dashboard**

- Uploaded syllabi
- Revision uploads
- Submission tracking
- Notifications

### **Student Dashboard**

- Approved syllabus repository
  - Search and filtering tools
  - Download access
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## **FUNCTION 3 — Syllabus Upload and Management**

## Who can upload

- Faculty Members
- Department Heads

## Upload Process

1. User opens upload form
2. User enters syllabus metadata
3. User uploads syllabus file
4. User clicks Save Draft or Submit

## Required Metadata

- Course Code
- Course Title
- Department
- Academic Year
- Semester
- Instructor Name

## Supported File Formats

- PDF

## Status Types

- Draft
- Submitted
- Approved
- Rejected

## Editing Rules

- Draft syllabi may be edited anytime
  - Submitted syllabi cannot be edited until reviewed
  - Rejected syllabi may be revised and resubmitted
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# FUNCTION 4 — Upload-Based Revision Tracking

## Purpose

Tracks uploaded syllabus revisions while preserving previous uploads for reference.

## How Revision Tracking Works

- Each new uploaded revision automatically creates a new version number
- Older uploads remain archived in the system
- Users may view previous uploaded versions

## Version Metadata

Each uploaded version stores:

- Version number
- Upload date and time
- Uploaded by
- Revision note
- File name
- Status

## Example Version Sequence

- v1 — Initial Upload
- v2 — Revised Upload
- v3 — Final Approved Upload

## Revision Notes

Users may optionally provide short descriptions of changes made during revision uploads.

## Standardized File Naming Convention

To maintain organized and consistent file storage, the system automatically renames uploaded syllabus files using a standardized naming format.

Filename Format:

**[CourseCode]\_[FamilyName]\_[Laboratory#/Activity#].pdf**

Example: **CPESD3\_LIMA\_LAB#1.pdf**

## **Version Comparison Viewer**

The system provides a simple version comparison feature that allows authorized users to visually compare current and previous syllabus versions using embedded PDF or image previews. Users may open two versions side-by-side within the web interface to manually inspect changes between uploaded files.

## **Embedded iFrame PDF Viewer**

The system uses an embedded iFrame-based PDF viewer to securely display syllabus files directly within the web application. Users can view approved syllabus documents without downloading the original file, ensuring secure and controlled access to syllabus content.

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# **FUNCTION 5 — Submission and Approval Workflow**

## **Stage 1 — Draft**

- User uploads syllabus
- Status remains Draft
- Only uploader and Department Head may view it

## **Stage 2 — Submitted**

- User submits syllabus for review
- Status changes to Submitted
- Department Head receives notification
- Uploader cannot edit while under review

## **Stage 3 — Department Head Review**

- Department Head reviews uploaded syllabus
- Department Head may:
  - Approve submission
  - Reject submission with remarks

## **Stage 4 — Approved**

- Status changes to Approved
- Syllabus becomes visible to students
- Uploader receives approval notification

## **Stage 5 — Rejected**

- Status changes to Rejected
  - Rejection remarks sent to uploader
  - User may revise and resubmit syllabus
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# **FUNCTION 6 — Advanced Search and Filtering**

## **Search Parameters**

Users may search syllabi using:

- Course Code
- Course Title
- Department
- Academic Year

- Semester
- Instructor Name

## **Role-Based Visibility**

- Students see Approved syllabi only
  - Faculty see their uploads and approved syllabi
  - Department Heads see departmental syllabi only
  - Admins access monitoring information only
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# **FUNCTION 7 — Activity Logging and Audit Trail**

## **Purpose**

Maintains a permanent record of important system activities.

## **Logged Activities**

### **Authentication**

- Login attempts
- Logout events
- Password resets
- Failed logins

### **Syllabus Activities**

- Upload syllabus
- Submit syllabus
- Approve syllabus
- Reject syllabus
- Upload revision

### **Account Activities**

- Registration approval
- Registration rejection



- Role updates
- Account deactivation

## **Audit Entry Information**

- Action type
  - Performed by
  - Date and time
  - Result status
  - Description
- 

# **FUNCTION 8 — Notification System**

## **Purpose**

Provides real-time updates regarding important workflow activities.

## **Notification Triggers**

### **Registration Approval**

Recipient:

- Newly approved user

### **Registration Rejection**

Recipient:

- Rejected applicant

### **Syllabus Submission**

Recipient:

- Department Head

### **Syllabus Approval**

Recipient:

- Faculty uploader

## **Syllabus Rejection**

Recipient:

- Faculty uploader

## **Revision Upload**

Recipient:

- Department Head

## **Notification Delivery**

- In-app notification center
  - Optional email notifications
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# **FUNCTION 9 — Email-Based Password Recovery**

## **Purpose**

Allows users to securely recover forgotten passwords.

## **Process**

1. User selects “Forgot Password”
2. User enters registered email address
3. System generates secure reset token
4. Reset link or code is sent to the user’s email
5. User creates a new password
6. Reset token expires after a limited time

## **Security Rules**

- Reset tokens are single-use only
- Expired tokens become invalid
- Password reset attempts are logged